Chapter 10, Problem 94P

Problem

Design a Colpitts oscillator that will operate at 50 kHz.

Step-by-step solution

Step 1 of 7

Consider the following data:

The operating frequency f_o is 50 kHz

The following are the conditions to design a Colpitts oscillator.

$R_i \gg X_{C_2}$

Assume the following data:

The values of the capacitors C_1 and C_2 are 20 nF.

Step 2 of 7

Calculate the capacitance C_T.

$$C_T = \frac{C_1 C_2}{C_1 + C_2}$$

Substitute 20 nF for C_1 and C_2 in $C_T = \frac{C_1 C_2}{C_1 + C_2}$

$$\begin{aligned} C_T &= \frac{(20)(20)}{20+20} \\ &= 10\text{ nF} \end{aligned}$$

Step 3 of 7

Calculate the value of the inductor L:

$$f_o = \frac{1}{2\pi\sqrt{LC_T}}$$

$$LC_T = \frac{1}{(2\pi f_o)^2}$$

$$L = \frac{1}{C_T(2\pi f_o)^2}$$

Step 4 of 7

Substitute 10 nF for C_T and 50 kHz for f_o in $L=\frac{1}{C_T(2\pi f_o)^2}$

$$\begin{aligned}L&=\frac{1}{(10\times10^{-9})\left[2\pi(50\times10^3)\right]^2}\\&=\frac{1}{(10\times10^{-9})(98.7\times10^9)}\\&=1.013\times10^{-3}\\&=1.013\text{ mH}\end{aligned}$$

Therefore, the value of the inductor L is 1.013 mH.

Step 5 of 7

Calculate X_{C_2}

$$\begin{aligned}X_{C_2}&=\frac{1}{\omega_o C_2}\\&=\frac{1}{2\pi f_o C_2}\end{aligned}$$

Substitute 20 nF for C_2 and 50 kHz for f_o in $X_{C_2}=\frac{1}{2\pi f_o C_2}$

$$\begin{aligned}X_{C_2}&=\frac{1}{2\pi(50\times10^3)(20\times10^{-9})}\\&=159.2\ \Omega\end{aligned}$$

Step 6 of 7

The value of R_i should be greater than X_{C_2} i.e. $R_i\gg X_{C_2}$.

Consider the value of R_i is $20\text{ k}\Omega$

For better gain the value of the resistor R_f should be greater than or equal to R_i.

Consider the value of R_f is $200\text{ k}\Omega$

Step 7 of 7

Calculate the gain of the Colpitts oscillator.

$$\frac{\mathbf{V}_o}{\mathbf{V}_i} = 1 + \frac{R_f}{R_i}$$

Substitute $20\ \text{k}\Omega$ for R_i and $200\ \text{k}\Omega$ for R_f in $\frac{\mathbf{V}_o}{\mathbf{V}_i} = 1 + \frac{R_f}{R_i}$.

$$\begin{aligned}\frac{\mathbf{V}_o}{\mathbf{V}_i} &= 1 + \frac{200 \times 10^3}{20 \times 10^3} \\ &= 1 + 10 \\ &= 11\end{aligned}$$

The gain of the Colpitts oscillator is 11.

Draw the designed Colpitts oscillator.

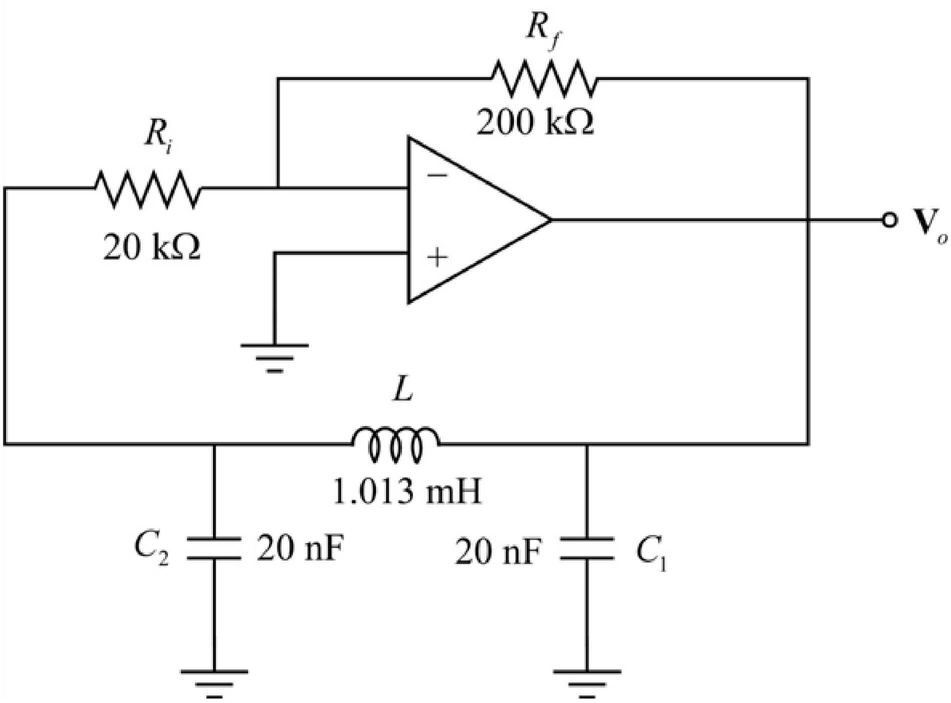

Figure 1